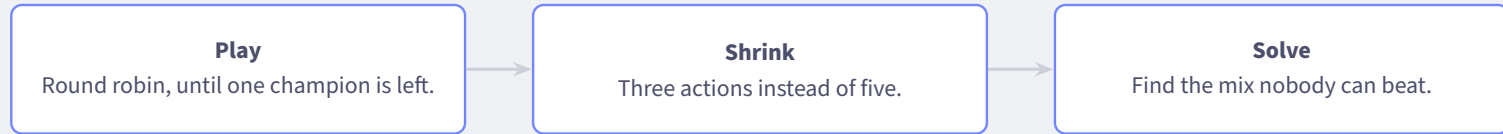


Nash equilibrium

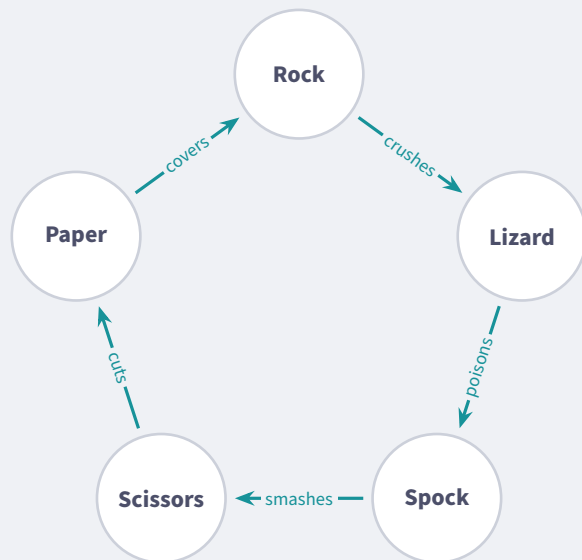
Rock Paper Scissors Lizard Spock, all the way down



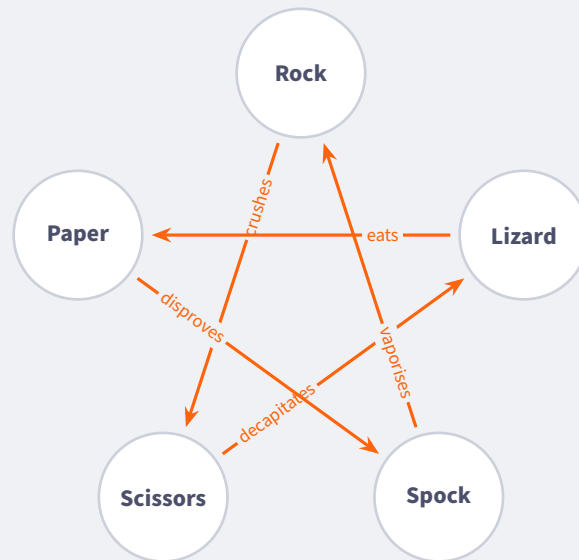
Class champion:

Did I beat them?

Beat the one beside you



Beat the one across from you



Each action wins , loses and draws . So the game is _____, and every cell's payoffs add to .

Action	Tally	Count	Share
Rock			
Paper			
Scissors			
Lizard			
Spock			

If everyone had played the same action, what should you have played?

	Rock	Paper	Scissors
Rock			
Paper			
Scissors			

Fill in the *row* player's payoff:

win = +1, draw = 0, lose = -1.

Then circle each player's best responses.

The column player's matrix is .

A cell with both players' best responses circled would be a pure Nash equilibrium.

If they play Rock, I want

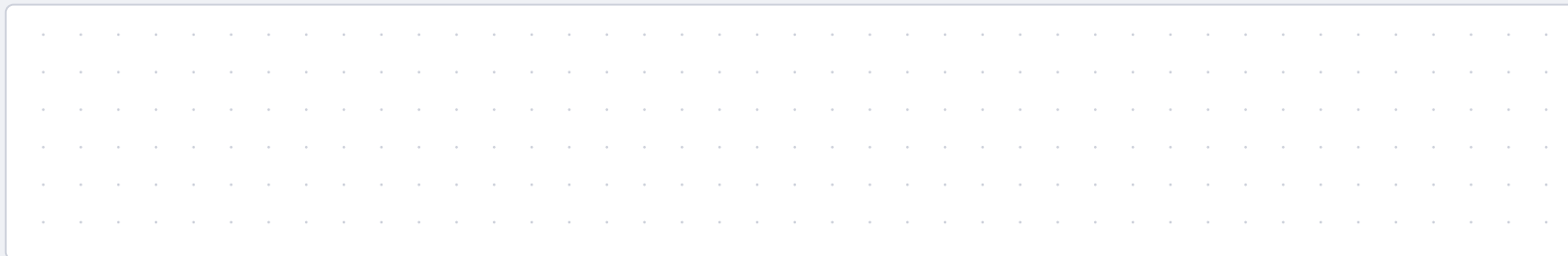
If they play Paper, I want

If they play Scissors, I want

... and then they want to move too.

So any equilibrium has to be **_____**.

A mixed strategy σ is a best response to τ if and only if



In words: every action I actually use has to be _____.

So if I mix over all three actions, all three must give me the same expected payoff.

$$u_r(\text{Rock}, \sigma_c) =$$
$$u_r(\text{Paper}, \sigma_c) =$$
$$u_r(\text{Scissors}, \sigma_c) =$$

Set them equal, and solve:

$$\sigma_c =$$

Support enumeration checks every pair of supports in turn.

Non-empty supports for one player, out of 3 actions:

Pairs of supports to check:

And with 5 actions, as in the game we played:

Which is why we want theorems, not enumeration.

What we just did in the room is **Question 1** on the Nash Equilibrium page.

- (a) Why it is symmetric and zero sum, and the column player's matrix. [3]
- (b) Utilities for three given pairs of mixed strategies. [3]
- (c) Use the best response condition to prove $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is an equilibrium. [5]
- (d) Prove it is the *unique* equilibrium. [5]
- (e) State and justify the equilibrium for the five-action game. [3]

vknights.org/gt/topics/nash-equilibrium.html